Higher National Diploma in Engineering (Electrical)

First Year, Second Semester Examination - 2021

EE 1209-Electronics Circuits and Systems- I

Answer any 5 questions Time : 3 hours

Question 1

(i) List the applications of diodes (2 marks)

(ii) Explain the **drift current** for semiconductors (3 marks)

(iii) Briefly explain the differences among an **Insulator, Conductor, and Semiconductor** with support of the **energy band** (4 marks)

(iv) Explain the **formation of a PN junction diode** (5 marks)

(v) Describe the **PN junction diode's forward and reverse bias characteristics** by referring necessary curve (6 marks)

Question 2

(i) What is a **capacitance diode**? (2 marks)

(ii) Explain the requirement of a **current limiting resistor** for **Zenor diode circuits** (3 marks)

(iii) Describe the **avalanche effect** (4 marks)

(iv) Explain the operation of the **full wave rectifier** with relevant circuit, graphs (5 marks)

(v) A **full wave rectifier** is used to energize 15V d.c. to a pure resistive load having 1kΩ and the diode has a forward resistance of 25Ω. Estimate the **peak value of the ac voltage** required at the input side (6 marks)

Question 3

(i) What is the **lead-acid accumulator** and its **Ah**? (2 marks)

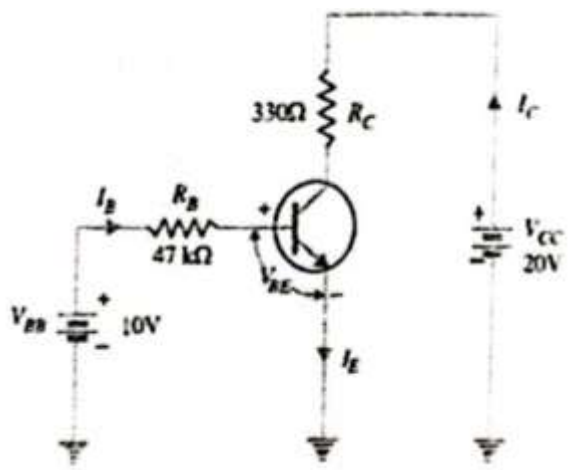
(ii) What are various **configurations associated with amplifier circuits**? (3 marks)

(iii) Draw **C filter and LC filter circuits** and explain the operation (4 marks)

(iv) a. Sketch the **PNP and NPN transistor symbols** (2 marks)

b. Describes the **operation of an NPN transistor** (3 marks)

(v) Determine the **base, collector, and emitter currents** and V_{CE} for the **common emitter amplifier circuit** shown in Figure 3(v). Where V_{BE}=0.7V, β=150 (6 marks)



Question 4

(i) When can a **transistor be used as a switch**? (2 marks)

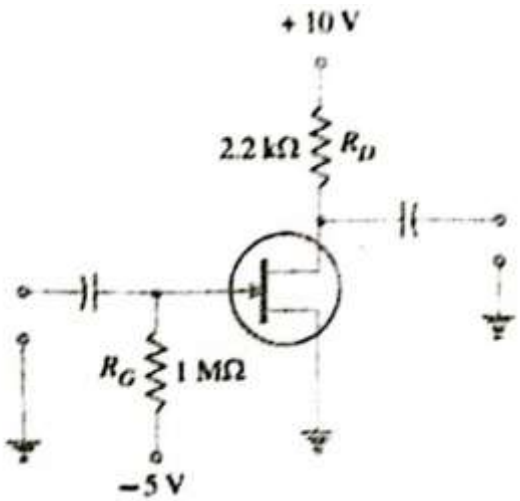
(ii) a. Why is **'FET' referred to as a "voltage-controlled device"**? (2 marks)

b. Describe the **operating principles of a junction field effect transistor (N-channel type)** (3 marks)

(iii) Derive an expression for **trans-conductance for JFET** (3 marks)

(iv) What is **MOSFET** and briefly explain **N-channel depletion type MOSFET** (4 marks)

(v) A **fixed biased JFET amplifier** having parameters I_{DSS} =20mA & $V_{GS(Off)}$=-8V is shown in Figure 4(V). Determine V_{GS}, I_D, and V_{DS}(6 marks)



Question 5

- (i) Convert the following **hexadecimal numbers to binary** (1 marks)
  - a. $27BC.D8_{16}$
- (ii) Convert the following **octal number to a hexadecimal number** (4 marks)
  - a. 1452_8
  - b. 367_8
- (iii) a. Explain the rules of adding two **BCD numbers** (2 marks)
  - b. Add $0101 + 1001$ by using the **BCD number** (3 marks)
- (iv) a. What is a **combinational logic circuit**? (2 marks)
  - b. State why combinational logic circuits differ from the **sequential logic circuit**? (3 marks)
- (v) a. Find the **2's complement and 1's complement of 101101** (2 marks)
  - b. **Subtract** 1000011_2 **from** 1010100_2 **using 2's complement method** and show the direct subtraction for comparison (3 marks)

Question 6

(i) State the **Commutative Property and Associative Property of Boolean algebra** (3 marks)

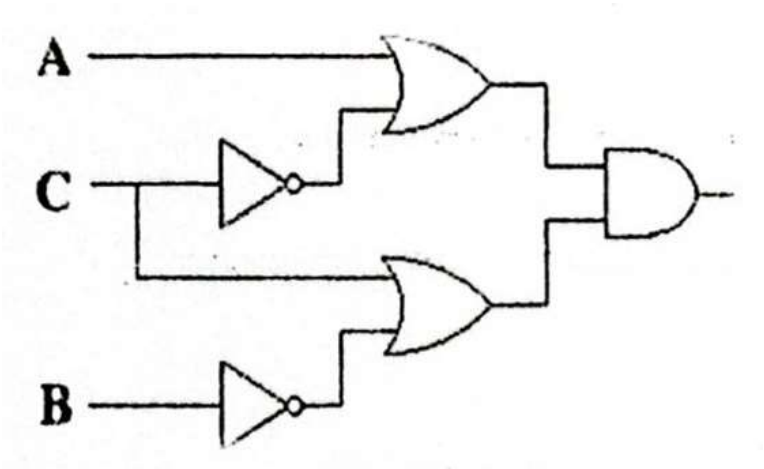
(ii) By using a **lamp and switches draw an electrical equivalent circuit** for (3 marks)

a. NOT gate
b. AND gate
c. OR gate

(iii) States **de Morgan's theorems** for two logic variables A and B (4 marks)

(iv) A logic circuit shown in Figure 6(iv) consists of three inputs A, B, C, and the output is X. (4 marks)

a. Write a **Boolean expression for output**
b. Draw the **truth table**



(v) Use a **Karnaugh map to simplify** the following (6 marks)

a. X=AC' + A'B + ABC

b. F(w,x,y,z)=∑(0,1,3,7,9,10,11,12,13)

Higher National Diploma in Engineering (Electrical)

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EE1208 - Electrical Principles B

Answer any 5 questions Time : 3 hours

Question 1

a.) Define the following terms related to graph of a network (6 marks)

- i.) **Tree**
- ii.) **Twigs**
- iii.) **Links**

b.) Consider the following circuit shown in the Figure 1 and answer the questions given below . Branches of this circuit is named as 1, 2, 3, 4, 5 and node voltages are V_A, V_B, and V_C respectively. Node 4 (V_D) is the reference node.

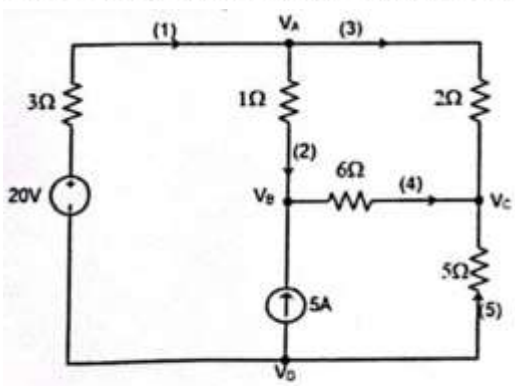


Figure1: Circuit for the Q1

- i.) Draw the **graph for the network** (2 marks)
- ii.) If the branches 2, 3 and 5 considered as the **twigs** of the graph, write the **fundamental tie-set matrix ([B])** for this graph and network (2 marks)
- iii.) What is the **fundamental cut-set matrix ([Q])** for this network (2 marks)
- iv.) Write the **branch impedance matrix** ($[Z_b]$) for this network or graph (2 marks)

c.) Apply the **nodal analysis** to the network shown in Figure 1 and calculate the **node voltages** V_A, V_B **and** V_C (6 marks)

Question 2

a.) Consider the following **T-network** shown in the Figure 2 and answer the questions.

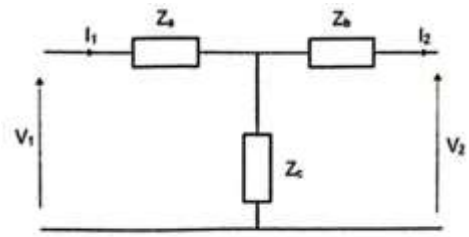


Figure 3: T-network for Q2. a)

i.) Calculate the **impedance parameters** for the above network in terms of Z_a, Z_b and Z_c

ii.) Calculate the **ABCD parameters** of the network in terms of Z_a, Z_b and Z_c (5 marks total for a. i. and a. ii.)

b.) Calculate the **impedance parameters and ABCD parameters** for following circuit shown in the Figure 3 (8 marks)

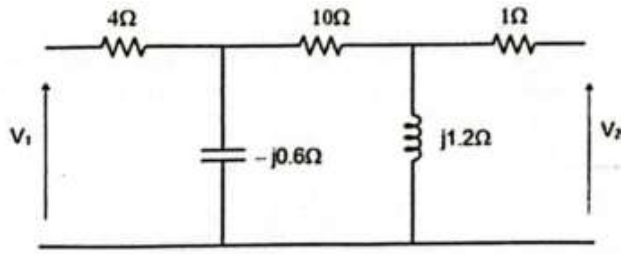


Figure 3: Circuit for Q2. b)

c.)

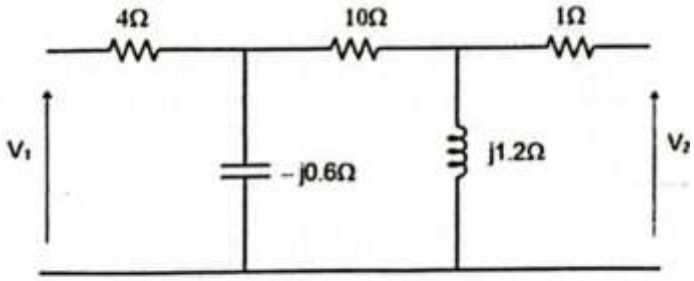


Figure 3: Circuit for Q2. b)

i.) Draw the basic building block passive **T-networks for high pass filter** (2 marks)

ii.) A **high pass T - section filter** having a cut-off frequency of 5kHz. Load should provide with 360V and 0.8A current. Determine the **values of the components required** and draw the complete circuit including the components and its values (5 marks)

Question 3

a.) Voltage variation and current variations of an incorrectly terminated transmission line is expressed using the following two equations:

$$\text{Voltage} : V(x) = (\frac{V_s - Z_0 I_s}{2})e^{\gamma x} + (\frac{V_s + Z_0 I_s}{2})e^{-\gamma x}$$

$$\text{Current} : I(x) = (\frac{I_s - {}^{V_s}/Z_0}{2})e^{\gamma x} + (\frac{I_s + {}^{V_s}/Z_0}{2})e^{-\gamma x}$$

Here, V_s is the sending end voltage and I_s is the sending end current.

- i.) Write the expressions for **voltage variation and current variation of a correctly terminated transmission line** (2 marks)
- ii.) Briefly explain how the **propagation coefficient of a transmission line effect the transmission of signals** (3 marks)
- iii.) What are the **three factors effect the distortions of the signals receive at the receiving end** (3 marks)

b.) An open wire transmission line is $120\ km$ **long** and it is terminated with its **characteristic impedance**. At the sending end is a generator having open circuit voltage of $10V$ and it has internal resistance of $360\ \Omega$. Supply frequency is $2\ kHz$. Primary constants of the line are: $R = 7.2\ \Omega/loop\ km$, $L = 3\ mH/loop\ km$, $C = 7.1\ nF/km$, and $G = 0.25\ \mu S/km$. Calculate:

- i.) **Characteristic impedance** of the transmission line (3 marks)
- ii.) **Propagation coefficient** of the line (2 marks)
- iii.) **Receiving end current** of the line (4 marks)
- iv.) **Speed of transmission** (3 marks)

Question 4: Transients and Magnetic Circuits

a.) Consider the following circuit shown in the Figure 4 . This circuit consists of a SPDT switch S and initially it is in the open position and at $t = 0$, this S switch is closed. If the capacitor is charged for $20V$, find the expressions for **current through the inductor and capacitor voltage** (8 marks)

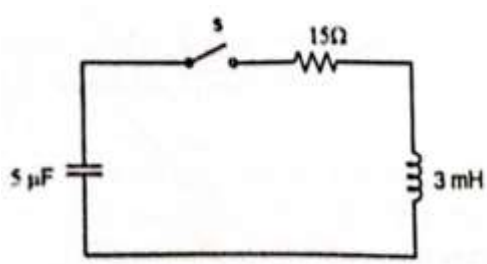


Figure 4: Circuit for Q4. a)

b.) i.) What are the **two types of transients in the power system** (2 marks)

- ii.) Briefly explain the importance of **Basic Impulse Level (BIL)** of any equipment or cable (3 marks)

c.) An **electromagnetic relay** is shown in the Figure 5 and it has an exciting coil of 1000 turns. The coil has a cross-sectional area of $2\ cm \times 2\ cm$. Neglect reluctance of the magnetic circuit and effect of fringing and find the following:

- i.) Find the **coil inductance** if the air-gap length is $0.6\ cm$ (3 marks)
- ii.) If the **flux density** in the air gap need to be maintained at $0.8\ T$ when the airgap length is $0.5\ cm$, find the **required current through the coil** (4 marks)

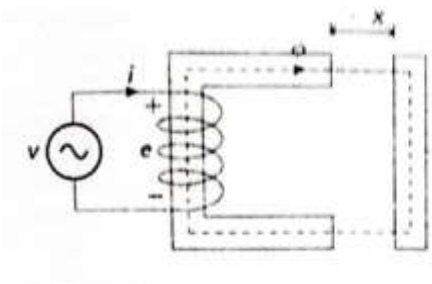


Figure 5: Magnetic circuit for Q4

Question 5

a.) i.) What are the **properties of electromagnetic waves** (2 marks)

- ii.) How the **Poynting Vector is calculated** and what information we can obtained from that (2 marks)
- iii.) The electrical field and magnetic field variation of a monochromatic electromagnetic plane wave is given by:

$$E = E_0 \, cos(kz - \omega t)\hat{y}$$

$$B = B_0 \, cos(kz - \omega t)\hat{x}$$

Derive an equation for **average power propagate by this wave** (3 marks)

- iv.) Electromagnetic wave propagates through **free space with** $3 \times 10^8 \; m/s$ **speed** and enter some other medium with **relative permittivity of** 0.6 **and relative permeability of** 9. Find the **speed of the wave in this new medium** (3 marks)

b.) A plane electromagnetic wave propagates in a material with **relative permeability** 2 is given by:

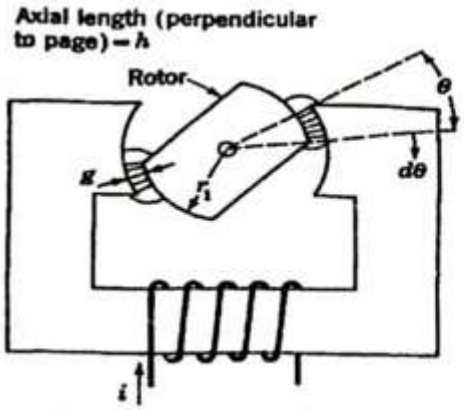
$$E = 50 \, cos(8x - 2.4 \times 10^8 t)\hat{y}$$

- i.) What is the **direction of propagation** of the wave (1 mark)
- ii.) Find the **wave length and frequency** of the wave (2 marks)
- iii.) Find the **velocity of propagation** in the medium (1 mark)
- iv.) Find the **magnetic field variation** (3 marks)
- v.) Find the **average power carried by the wave** (3 marks)

Question 6

a.) i.) Write the **energy balance equation for motor action** and derive the **power balance equation** (2 marks)

- ii.) Write the **power balance equation for following single exciter system** shown in Figure 6 and show that the **torque equation for the motor is given by** $T = \frac{1}{2} i^2 \frac{dL}{d\theta}$ (4 marks)



A singly excited rotating actuator

Figure 6: single exciter rotary system

- iii.) Briefly describe the terms **stored energy and co-energy** (2 marks)

b.) Draw the typical **magnetization of core materials** use in electrical equipment and explain the **importance of this characteristics in machine design** (3 marks)

c.) i.) State **four advantages of fiber optic communication** over other communications medias (4 marks)

- ii.) **Fiber optic cable** is used to transmit some data and the cable core diameter is $50\ \mu m$ and the light use for the transmission has wave-length of $770\ nm$. Refractive index of the core and cladding are 1.48 and 1.45 respectively. Calculate the **number of modes of the optical fiber** (5 marks)